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CLOUD DATA ENGINEERING:
MICROSOFT FABRIC

Deployment

By: Ahmed Irshad Hussain



Data Engineering: Microsoft Fabric Deployment

DEPLOYMENT (Concept First)

What I built so far

CSV → Bronze → Silver → Gold → Power BI

But right now:

I manually running notebook

What “Deployment” means in My project

Deployment is NOT:

- Hosting a website
- Complex DevOps

In data engineering, deployment means:

Automating your pipeline so it runs without you

My pipeline after deployment

New Data → Pipeline Runs Automatically → Tables Update → Dashboard Updates

Final Understanding

Deployment =

Manual system → Automated system

Microsoft Fabric

STEP 1: Create a Pipeline

- Do this: Go to your workspace
- Click New Select & select Data Pipeline
- Name it: Any

(I'll give it "AhmedSales_pipeline_automation")

The screenshot shows the Microsoft Fabric interface for a workspace named 'Fabric Ahmed'. At the top, there is a '+ New item' button highlighted with a red box. Below it is a table of items:

	Name	Status	Type	Task
	Ahmed_PowerBI_1		Semantic ...	—
	Ahmed_Sales_Lakehouse		Lakehouse	—
	Ahmed_Sales_Lakehouse		SQL analyt...	—
	Sales Dashboard		Report	—

A red arrow points from the 'Sales Dashboard' row to a selection menu. In this menu, the 'Pipeline' option is highlighted with a red box. Another red arrow points from the 'Pipeline' option to the 'New Pipeline' dialog box.

The 'New Pipeline' dialog box has the following fields:

- Name ***: AhmedSales_pipeline_automation
- Location ***: Fabric Ahmed (with a dropdown arrow and an external link icon)
- Create** and **Cancel** buttons at the bottom right.

Home

Workspaces

Copilot

OneLake catalog

Fabric Ahmed

AhmedSales_pipeline_a...

My workspace

...

Fabric

Fabric

AhmedSales_pipeline_automation X

Search

Trials activated: 31 days left

22

?

Home

Activities

Run

View

Validate

Run

Schedule

Trigger

View run history

Copy data

Dataflow

Notebook

Transform

Spark Job Definition

Notebook

Script

Select this one!

Build a pipeline to organize and move your data



Pipeline activity

Automate data orchestrations using rich no-code activities.

Copy data assistant

Follow guided steps to copy data into Microsoft Fabric, as well as other data stores.

Practice with sample data

Quickly build a pipeline with a predefined template to load data into Lakehouse.

Templates

Generate a new pipeline quickly using a predefined data scenario.

Need help? Ask Copilot

Microsoft Fabric

STEP 2: Connect the notebook:

- Go to the settings of the pipeline
- Select your notebook
- Make sure to select the medallion architecture notebook.

STEP 3: Run the notebook

After Successfully automated the pipeline, now I have:
Notebook → Pipeline → Automated Execution

In Microsoft Fabric terms:

- I built ETL logic
- I orchestrated it
- I executed it via pipeline

The screenshot displays the Microsoft Fabric interface. At the top, a 'Notebook' card for 'Ahmed_Automation' is visible. Below it, the 'Settings' tab is active, showing configuration options for 'Connection', 'Workspace' (Fabric Ahmed), and 'Notebook' (Transformation&Cleaning). A red box highlights the 'Notebook' dropdown, with a red arrow pointing down to the 'Run' button in the 'Run' tab. The 'Run' button is also highlighted with a red box. Below the 'Run' button, a red arrow points down to the 'Output' tab, which shows the pipeline execution results. The 'Output' tab displays the 'Pipeline run ID' (63a62c38-380e-4214-aaf4-f9af9c6fc56a), the 'Pipeline status' (Succeeded), and a table of activity results.

See the status,
The pipeline ran successfully

Activity name	Activity status	Run start	Duration
Ahmed_Automation	✓ Succeeded	5/6/2026, 12:47:22 AM	3m 20s

Microsoft Fabric

STEP 4: Add Schedule:

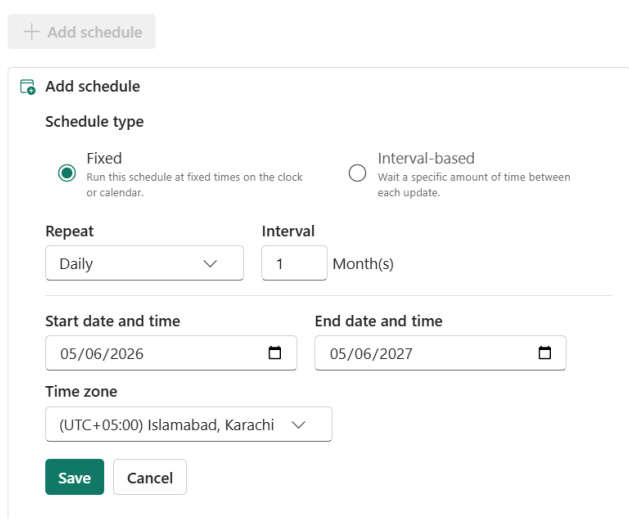
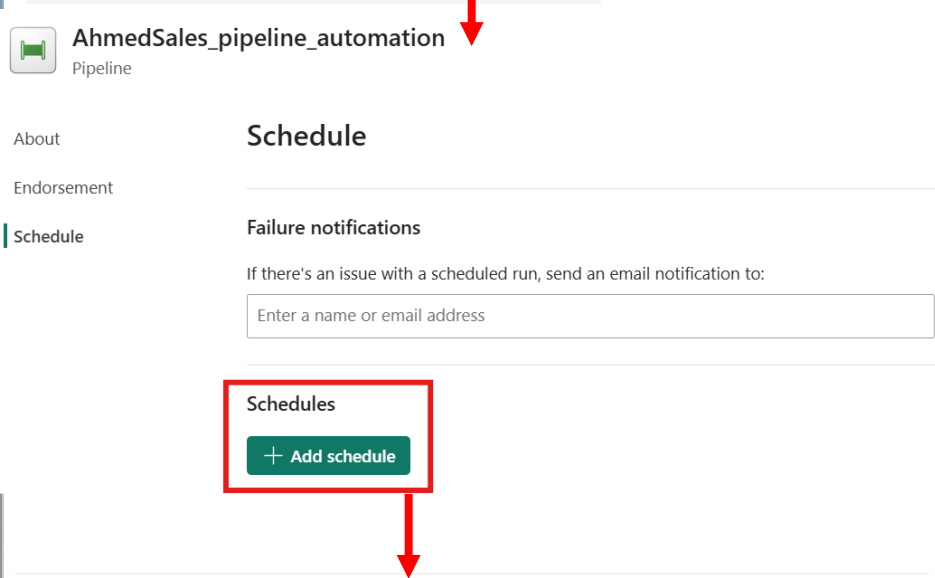
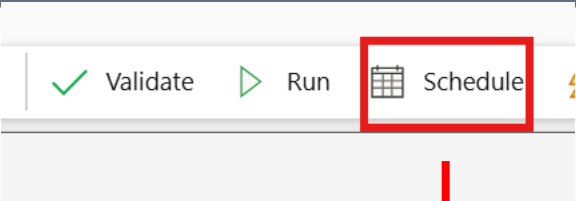
- Right now, it runs manually
- We need : Runs automatically WITHOUT me!

What to Do :

1. Click **Schedule** (top bar)
2. Click **New Schedule**

STEP 5: Setup the Schedule:

- Select schedule type
- Then select repetition (hourly, daily, monthly), I selected daily
- Set date and time
- Set time zone



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Schedule Type → Fixed
Repeat → Daily
Interval → 1
Time → (e.g. 8:00 PM)
Time Zone → Pakistan (UTC+05:00)



About

Endorsement

Schedule

Schedule

Failure notifications

If there's an issue with a scheduled run, send an email notification to:

sardarahmed.irshad2@gmail.com *

Schedules



Every day



On

Time of day	Last successful update	Next update
20:00	Not available ⓘ	18 hour(s) 51 minute(s)

Start	End	Time zone
06/05/2026	06/05/2027	(UTC+05:00) Islamabad, Karachi

Job type	Modified By
Pipeline	Ahmed Fabric ⓘ

^ Show less

ID: 47ab438f-5974-47af-8cbb-0b209d7bfa4e



Run



Edit

+ Add schedule

Microsoft Fabric

STEP 6: Power BI auto refresh:

- Right now:
- Pipeline → runs automatically
- Dashboard → still manual refresh

Let's fix that,
Go back to Semantic Model
Click → Schedule refresh
Configure it

STEP 7: Setup the Schedule:

- Set time zone.
- Set frequency to daily and time, 15 to 30 minutes ahead of pipeline schedule.

Semantic model
Ahmed_Power

Overview Lineage Mo

Make this item more discoverable by adding a description. Just start typing...

Location	Refreshed	Owner
Fabric Ahmed	02/05/26, 18:48:57	Ahmed Fabric

Refresh

Keep your Direct Lake data up to date

Configure Power BI to detect changes to the data in OneLake and automatically update the Direct Lake tables that are included in this semantic model. [Learn more](#)

☒ On

Time zone

Time zone configuration is applied not only to determine the schedule refresh time but also to establish the current date and time for incremental refresh models during on-demand and API refreshes. [Learn more](#)

(UTC+05:00) Islamabad, Karachi

Configure a refresh schedule

Define a data refresh schedule to import data from the data source into the semantic model. [Learn more](#)

☒ On

Refresh frequency

Daily

Time

8 30 PM

[Add another time](#)

Send refresh failure and critical warning notifications to

☒ Semantic model owner

☐ These contacts:

Enter email addresses

Query Caching

Refresh

Server settings

I set dashboard refresh time, 8: 30pm

- Pipeline → 8:00 pm
- Refresh → 8:30 pm

Why delay?

- Because: Pipeline must finish first THEN refresh dashboard should run

Pipeline run details



Owner

Ahmed Fabric

Submitted by

Ahmed Fabric

Start time

5/7/2026, 8:00:01 PM

End time

5/7/2026, 8:04:58 PM

Status

 Succeeded

Parameters

0

Pipeline run ID

c421833d-9f62-42b5-b82b-249fce5cfe12



Semantic model

Ahmed_PowerBI_1

Overview

Lineage

Monitor

Permissions

Make this item more discoverable by adding a description. Just start typing...

Location

Fabric Ahmed

Refreshed

07/05/26, 20:34:29

Owner

Ahmed Fabric

As you can see now,
Pipeline auto refreshed at 8:00 pm and semantic model at 8:30pm (30minutes ahead to the pipeline schedule)

Time, when I took
Screen Shot

12:52 AM
5/8/2026 

Step 1: Created Pipeline

- New → Data Pipeline
- Name: AhmedSales_pipeline_automation

Step 2: Added Notebook

- Added Notebook activity
- Selected ETL notebook (Bronze → Silver → Gold)

Step 3: Test Run

- Executed pipeline
- Status: **Succeeded**

Step 4: Schedule

- Frequency: Daily
- Interval: 1
- Time: 8:00 PM

Step 5: Auto Refresh

- Enabled Semantic Model refresh
- Time: 8:30 PM

Final Flow

Pipeline → Bronze → Silver → Gold → Refresh → Dashboard

Outcome

- Automated pipeline
- No manual execution
- Live dashboard updates

FLOWCHART

